

Lab 8: Multi-Agent Investment Analysis with CrewAI

Lab Overview:

This is an example of multi-agent stock investment analysis that I developed in the lab using CrewAI. It has several specialized agents that work together to do stock analysis, collect data, assess risk, and provide suggestions on stock investment. In this lab, I was tasked with learning how to create a multi-agent collaborative workflow, delegation, and use of several technologies such as web scraping, financial calculations, and RAG in AI workflow. Four major agents were involved in this workflow – Market Research Analyst, Research Analyst, Trading Strategist, and Portfolio Manager.

Customization and Modifications:

Regarding the practical implementation of the lab, I have made some modifications to the investment crew itself. Specifically, I have changed the name and functionality of the strategic agent to Risk Aware Strategist instead of a standard trading agent. In addition, this agent concentrated more on entry price, stop loss, target price, and the risk reward balance than providing an easy buy or sell signal. The second modification that I have introduced is the introduction of a financial calculator that calculates the risk reward ratio, capital at risk, and suggested position size. This was meant to bring some structure and realism into the strategy, making it like the portfolio management process. Finally, I introduced live financial data using the yfinance library and a simple web scraping function from Yahoo Finance.

Stocks Tested:

The system was tested using the following stock symbols:

TSLA, GOOGL, NVDA

These companies were selected since they reflect different types of firms and market behavior. TSLA is more volatile and growth-oriented, GOOGL is more stable and fundamentals-oriented, and NVDA is growth-oriented due to AI and semiconductor demands.

Observation:

Upon testing the multi-agent system through all three stocks, it became apparent that each of the agents provided a unique analysis for each of the stocks tested. The Market Analyst looked at market trends and positioning, while the Research Analyst analyzed the fundamentals of the company and any risks associated with it. The Strategist developed strategies on the basis of positions and risks, while the Portfolio Manager formulated the final recommendations based on different viewpoints. In TSLA, the system made cautious recommendations owing to its volatility and high valuation issues. The case was not the same with GOOGL because the system recommended stable investments for the company with solid fundamentals and growth. In NVDA, the recommendations made were mostly positive considering its growth and strong market momentum because of the AI industry. A critical finding is that multi-agent systems generate more coherent and logical outputs as opposed to the single model since each agent has a particular focus. The final recommendation was well-organized because it had inputs such as market trend, fundamentals, and risk management.

Customization Summary Table:

Component	Original Lab Idea	My Customized Version
Strategic Agent	General trading/investment strategist	Risk-Aware Strategist focused on conservative decisions
External Data	General financial tools	Yahoo Finance scraping + yfinance metrics
Financial Tool	Optional calculator	Added risk_reward_calculator for risk/reward and position sizing
Workflow	Multi-agent CrewAI workflow	Sequential 4-agent workflow with focused responsibilities
Tested Stocks	Example stock analysis	TSLA, GOOGL, NVDA

Agent Role Comparison:

Agent	Purpose in My Notebook	Output Type
Market Research Analyst	Analyze stock trend, positioning, and market view using scraped stock data	Short market trend summary
Research Analyst	Analyze fundamentals, valuation, growth, and risks using stock metrics and scraped data	Short fundamental analysis
Risk-Aware Strategist	Create conservative strategy using entry, target, stop-loss, and risk/reward logic	Short trading strategy
Portfolio Manager	Combine all outputs into one final recommendation	Final BUY / HOLD / SELL recommendation

Stock Testing Comparison:

Stock	Expected Profile from Analysis	Main Observation
TSLA	Higher volatility, growth-focused	More cautious strategy due to risk and volatility
GOOGL	Stable, strong fundamentals	More balanced and fundamentals-driven recommendation
NVDA	Strong growth and momentum	More positive growth-oriented outlook

In the customized notebook, I modified the Strategic Agent to be called Risk-Aware Strategist and incorporated two primary improvements related to data collection, which included Stock Web Scraper and the incorporation of yfinance module for obtaining real-time metrics on stocks. Financial Calculator was included to calculate the risk/reward ratio and position sizing. Four special agents were applied for the reasoning process using the sequential CrewAI approach, which aided in structuring the results. The tests carried out for the stocks of TSLA, GOOGL, and NVDA revealed how flexible the system is regarding reasoning based on the company's characteristics, its volatility, and growth potential.

What I learned:

The following were some of the key learnings from this lab:

- The functioning of multi-agent systems when working together and delegating tasks
- The inclusion of tools in agents for fetching data externally
- The benefit of having real-time data for improved decision-making
- RAG usage for integration of external documents and information
- Sequential workflows for building upon agents' outputs
- Incorporation of risk management for better decision-making in investments

One key learning from this lab was about designing agent roles and task descriptions. In case they were not designed properly, agents could produce overlapping and confusing results.

Conclusion:

To conclude, the experiment demonstrated that multi-agent AI systems can be applied for decision-making on complex issues, such as analysis of investments. Since the division of different tasks was done using various agents, the result output from the system was well presented. Moreover, real-time data, financial tools, and risk analysis were utilized, making the whole process much more real and applicable. This experiment made it possible for me to get acquainted with CrewAI, agent delegation, tools usage, and multi-agent systems.

Completion Certificate:

Congratulations!

You Completed this Module

Advanced CrewAI: Multi-Agent Investment Analysis

Mastered agentic AI systems, delegation workflows, and autonomous AI agents for complex task execution.

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⌚ Time

300 min

🏆 Completed

April 06, 2026 at 05:02 PM

📚 Skills Mastered

- 💡 **Delegation:** Agents assign tasks to others
- 🔍 **RAG:** Retrieval augmented generation with documents
- ⚡ **Real-time:** Provides current and accurate information

🎉 **Congratulations** on mastering advanced AI agent collaboration!